Pannopi: a tool for pan-genome based prokaryotic genome assembly and annotation

PRESENTER:
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BACKGROUND: The volume of genomic data is growing rapidly. We must identify routine areas of analysis and automate them. For this purpose we have created Pannopi – a pipeline for large-scale prokaryotic genome analysis.

METHODS

1. Snakemake based
2. Includes 24 samples testing dataset
3. Uses most popular and effective tools
4. External scripts
5. Flexibility and customization

REPORT INCLUDES

- raw data and assembly quality control;
- cleaned from contamination assembly;
- structural annotation;
- pangenome-based comparative annotation.
- functional annotation with lists of antibiotic resistance and virulence genes, plasmids, phages, IS elements, tandem repeats, mlst-type, and serotype;

RESULTS

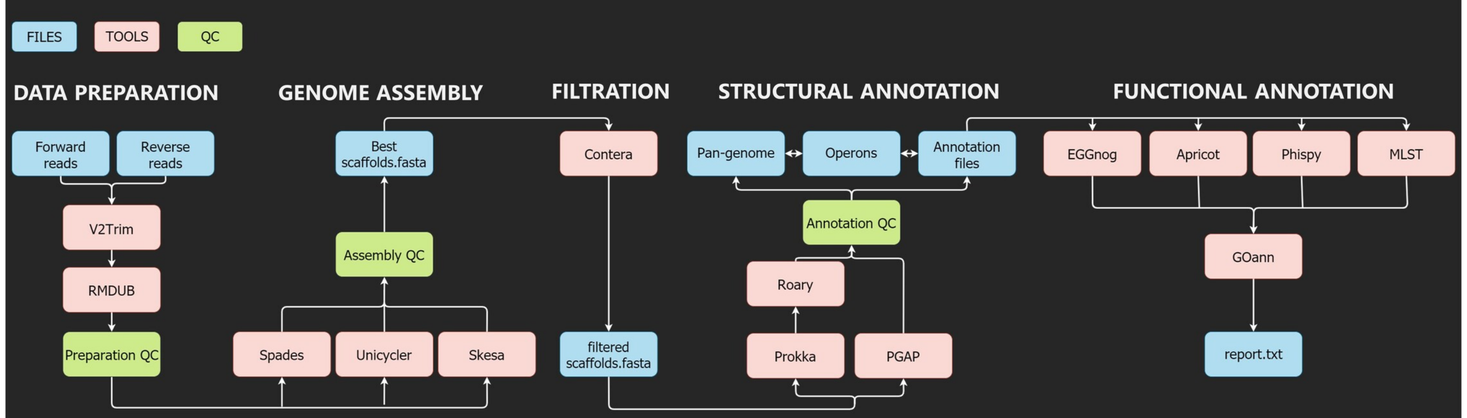
Analysis time of a single genome depending on the configuration of the pipeline.

Data preparation	Genome assembly	Filtration	Structural annotation	Functional annotation	**Full analysis time***
5 min	5-20 min	1 min	40-120 min	40 min	**91-186 min**

* Results obtained on system with 32 threads and 128 GB of RAM.

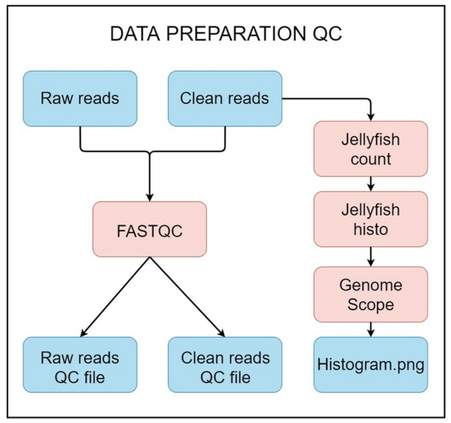
SCAMT

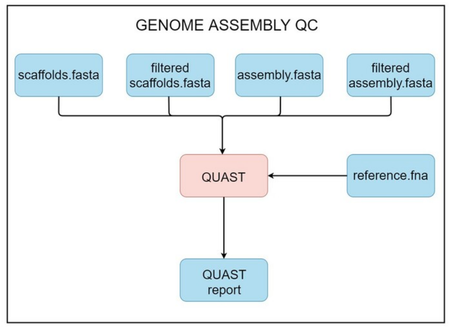
From reads to report **prokaryotic genome analysis** with **functional annotation** and **pan-genome construction**.

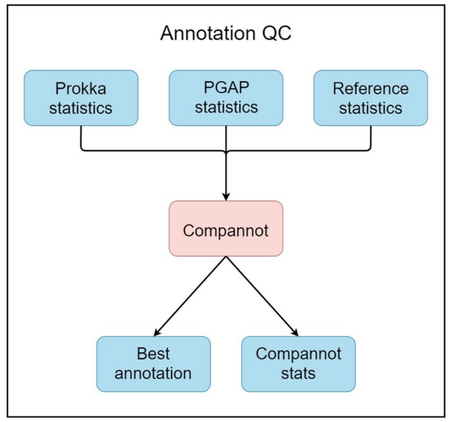


Take a picture to contact us

ADDITIONAL MATERIAL







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